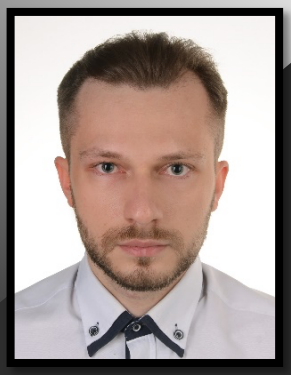

# Igor Slazyk

Machine Learning Engineer

## Portfolio

islazykv.github.io/portfolio

## LinkedIn

linkedin.com/in/igor-slazyk

## GitHub

github.com/islazykv

## Personal Information


| | |
|---|---|
| Nationality: | Polish |
| Date of Birth: | 03/11/1992 |
| Address: | Sandnes, 4337 Norway |
| Phone Number: | +47 [redacted] |
| | +48 [redacted] |
| E-mail: | islazykv@gmail.com |


## Profile

I am a positive and cheerful person who enjoys challenges. I approach every task with dedication and commitment, seeking effective solutions and paying attention to details. I am communicative and team-playing oriented, but I can also work independently and take initiative. I am open-minded, adapt quickly and I am always eager to learn.

## Interests

Popular Science Literature
IT / Technology
Gaming
Cycling
Hiking

## Summary


ML engineer with 8 years of experience in data science, including 4 years focused on machine learning. Passionate about turning data into valuable insights and impactful solutions. Driven to produce high-quality, scalable code and models. Collaborative team player, eager to contribute to projects at the intersection of AI and MLOps.


## Education

**Ph.D. in Particle Physics** — 10.2017 – 07.2021
University of Ferrara — Ferrara, Italy
Institute of Nuclear Physics — Krakow, Poland

**M.Sc. in Applied Physics** — 02.2016 – 05.2017
Krakow University of Technology — Krakow, Poland

**B.Sc. in Applied Physics** — 09.2012 – 02.2016
Krakow University of Technology — Krakow, Poland

**B.Sc. in Civil Engineering** — 09.2011 – 02.2015
Krakow University of Technology — Krakow, Poland

## Experience

**Postdoctoral Research Fellow** — 01.2022 – currently
Høgskulen på Vestlandet — Bergen, Norway

- Development of ML models for data analysis in high-energy physics
- Design and implementation of scalable end-to-end ML pipelines
- Data preprocessing, feature engineering, model evaluation and optimization
- Data analysis and interpretation of results from ML-based physics searches
- Code maintenance, refactoring and automated testing
- Preparation of publications and presentation of ML results

**Affiliated Researcher (ATLAS)** — 01.2022 – 03.2026
CERN — Geneve, Switzerland

- Software development within the ATLAS experiment framework (Athena)
- Development of TauCP software for tau particle reconstruction and identification
- Code optimization, refactoring and integration into large-scale workflows
- Evaluation and interpretation of physics analysis results
- Collaboration with developers, analysts and physicists on software and data tools

**Affiliated Researcher (LHCb)** — 01.2018 – 03.2021
CERN — Geneve, Switzerland

- Software development for automated quality assurance of RICH detector modules
- Data analysis of quality assurance results for elementary detector cells
- Development of data processing scripts and tools for measurement workflows
- Hardware assembly, testing and quality control measurements
- Initial exploration of ML approaches for detector data analysis

**Front-End Developer** — 10.2016 – 09.2017
Freelance — Cracow, Poland

- Development of responsive websites using React.js
- Collaboration with back-end developers on feature integration and functionality

**R&D Physics Intern** — 04.2016 – 09.2016
Camlin Technologies — Zurich, Switzerland

- Data analysis of optical and quantum measurements of semiconductor materials
- Data collection and processing for photoluminescence spectroscopy
- Collaboration with software developers and physicists on analysis workflows

## Projects

### Tau Supersymmetry Search

- repository
- notebook showcase
- publication

Boosted Decision Trees | Neural Networks

- Developed a supervised ML pipeline for multiclass classification with MLflow
- Trained and optimized classifiers with Optuna tuning
- Implemented ML-based regions construction and statistical hypothesis testing

### Tau Anomaly Detection

- repository
- notebook showcase

Autoencoder | Variational Autoencoder

- Developed an unsupervised ML pipeline for anomaly detection with W&B
- Trained and optimized models with Ray Tune tuning
- Implemented reconstruction error anomaly scoring and latent space analysis

### Tau Energy Scale

- internal project

Mixture Density Network | Neural Networks

- Developed a probabilistic regression model with uncertainty estimation
- Trained with Huber loss pretraining and NLL fine-tuning for density estimation
- Evaluated model performance with energy response and resolution histograms

### Sphalerons and Black Holes

- publication

Convolutional Neural Network | Boosted Decision Trees

- Optimized high-capacity models to classify rare high-multiplicity LHC events
- Developed a pipeline mapping detector signals into three-layer event images
- Compared high-level features with raw pixel data to maximize signal sensitivity

### Di-Tau Reconstruction

- internal project
- note

Software Development | Data Science

- Engineered a C++ analysis framework for boosted di-tau reconstruction studies
- Benchmarked fixed & variable radius subjets to optimize reconstruction efficiency
- Evaluated diverse seed jet collections to establish baseline configurations

### Elementary Cells

- repository
- publication

Software Development | Data Science

- Developed an automated C++/Python/LabVIEW software for photomultiplier QA
- Analyzed array performance via noise, dark count and threshold measurements
- Explored ML-based particle identification for the LHCb RICH detector

## Skills

**Programming Languages & Tools:**

- Python, C++, SQL, Bash
- Git, Jupyter, pytest, uv, conda

**Data Processing & Visualization:**

- NumPy, pandas, SciPy, ROOT, uproot, Awkward Array, pyhf
- Matplotlib, seaborn, Plotly

**Machine Learning & Deep Learning:**

- scikit-learn, XGBoost
- PyTorch, PyTorch Lightning, TensorFlow, Keras

**Tuning & Explainability:**

- Optuna, Ray Tune, SHAP

**Tracking & Deployment:**

- MLflow, W&B, DVC, Hydra
- FastAPI, Docker, GitHub Actions

**Web Development:**

- HTML, CSS, JavaScript, React

## Languages

| | |
|---|---|
| English: | Professional |
| Norwegian: | Basic |
| Italian: | Basic |
| Polish: | Native |

**Currently Learning:**

- Kubernetes
- Azure ML
- Azure DevOps Pipelines
- Databricks